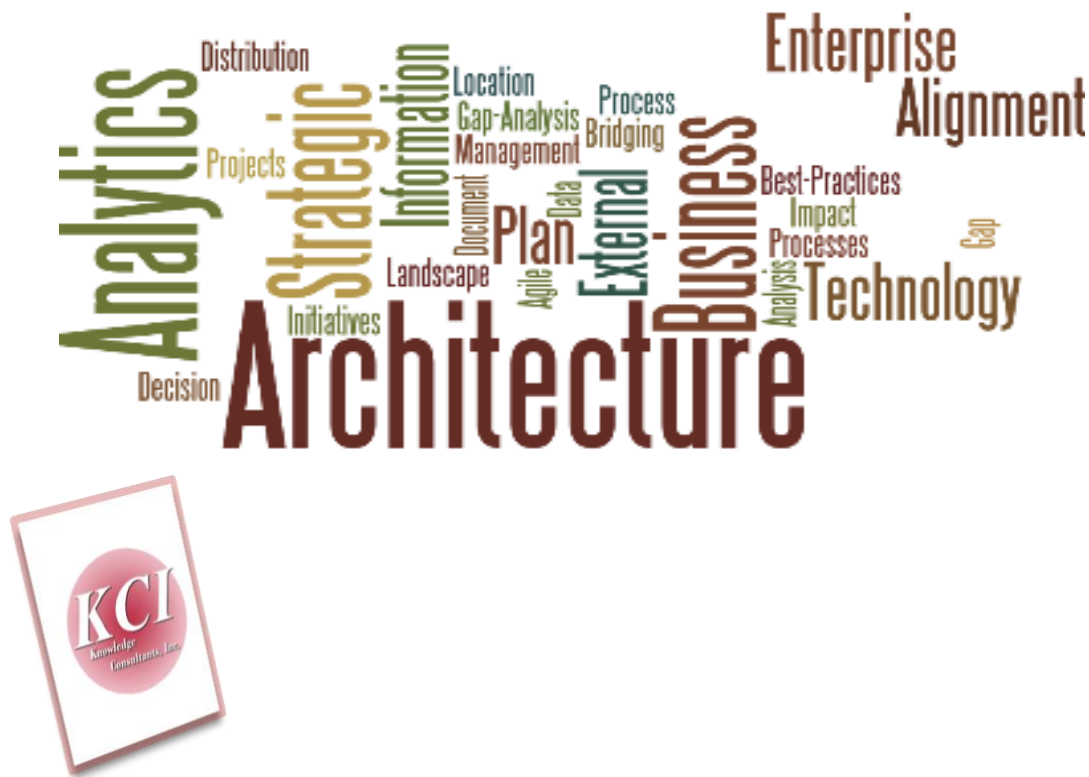


Knowledge Consultants, Inc.

BPM and Case Management Global Summit Washington, D.C. June 24, 2015 Workshop

Analysis & Analytics based on Architectures



Exercises we'll do:

We are
Here

Exercise 1 – Rank markets using the external environment defined by PESTLE – M

- Defines part of external architecture and links External-to-Strategic architectures

Exercise 2 – Identify which products are most impacted by the market rank and infer which major processes are impacted by the market rank

- Links Strategic-to-Business architectures

Exercise 3 – Infer which systems are impacted by the market rank

- Links Business architecture-to-Enterprise architecture

Setting the Stage

DAC (Dominion Appliance Company) was founded over 50 years ago by the grandfather of the current CEO. They have grown steadily over the many years the family has run the business and expanded into many countries around the world. DAC is chartered with providing appliances into the global marketplace. They have over 50 facilities around the world with distribution points on all major continents and manufacturing in Southeast Asia and the Middle East. Parts are supplied from five different locations in the world other than their own manufacturing facilities. To supply the global marketplace, they develop new products on a regular basis as well as upgrade existing products. Many of the products are basic to a household, such as: dishwashers, clothes washers, dryers and so on.

Situation and Background:

The CEO would like to take the company public in the next few years. He believes the company is not growing fast enough and will not attract investors. What he thinks would help is a systematic company improvement program that would make their performance attractive to outside interests such as another company to buy them, someone to merge with or a group of investors to take over and take the company public.

The CEO and several senior managers recently attended a strategic planning and business architecture conference and were impressed with the discussion about the value of business architecture. They were told that architecture is really about the structure of the business, its business components and how the components are linked together. With this material management could look at options and the impact of consolidations, new systems, addition and removal of products, mergers, acquisitions, divestitures and many other transformation alternatives that change the structure of the enterprise. In addition, these changes have significant

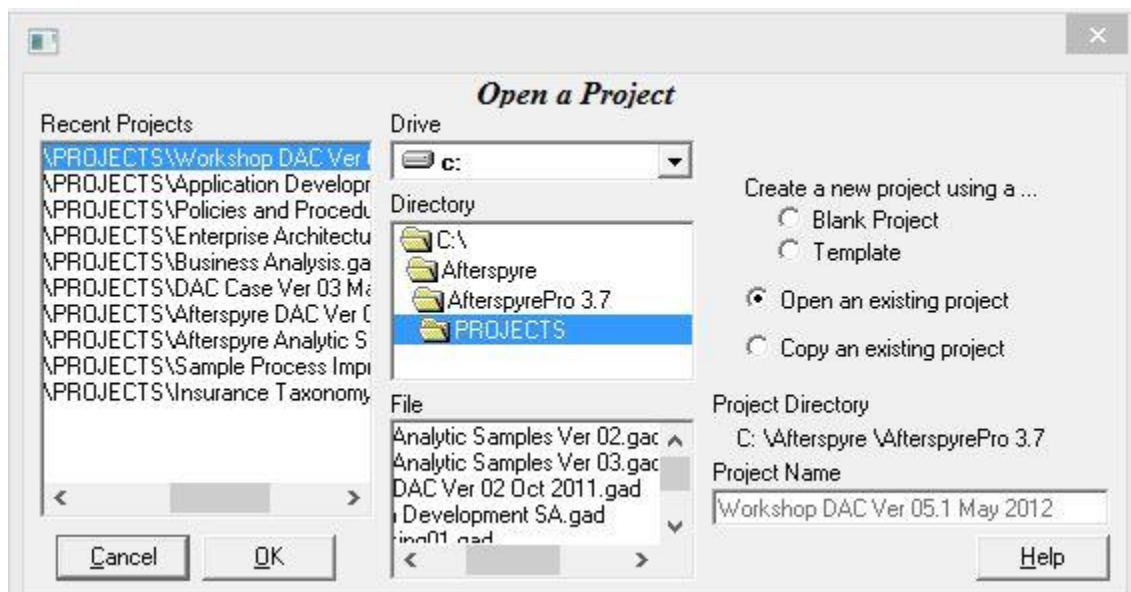
impact on the operational digital part of the business, its systems and the requirements for changes and new systems. Further changes in the digital part of the business may impact the capability to deliver to customers.

Management believes that several programs are needed to make the company perform better, such as: changes in the structure of the business with new strategies, business operation, performance management and several others. You and several others are assigned to a 'new directions' team and are given various assignments related to this company as you learn more about how the company operates.

Setting up the software:

We are using an older version of the architecture analytic engine called Gadrian.

1. Installation of Gadrian product that we will use for analytics.
 - a. The file name is **GadrianPro.msi 3.6.2**. Double click and the install will start. This will only install on Windows/Intel machines using Win XP, Win 7, Win 8 and above.
 - b. Follow the install instructions like any other products.
 - c. Go to the root folder on the C drive and find the **Gadrian** folder.
 - d. Open the **Gadrian 3.6** folder
 - e. Open the **Projects** folder
 - f. Copy a file named "**Workshop DAC Ver. 5.1 June 2015.gad**" file to the Projects file.
 - g. Go to Start, Programs, and select **GadrianPro**.
 - h. The Gadrian product should open to a selection screen for a project.



- i. Click on the Workshop project in the list box and then **OK**

2. In the upper left menu of Gadrian, there are four squares: **H** – Home, **B** – Browser, **A** - Analytics, and **U** – Utilities which can be used. We will use only the following 2 features for the exercises, they are the:
 - a. Model **B**rowser – Where you view and change models
 - b. Analytics workspace – Where the analytics are performed

What you have for the exercises:

Material is positioned in the Gadrian tool for use in the tool based exercises. For example there are:

1. PESTLE based models components such as
 - a. Markets
 - b. Technology
 - c. Social trends
 - d. Legislation via regulatory agencies
 - e. Economic factors
 - f. Environmental factors
 - g. Key matrices relating Markets to the PESTLE factors
2. There is also corresponding for each of the architecture perspectives. These materials include lists of:
 - a. Processes
 - b. Business Applications
 - c. Databases
 - d. Technology (IT)
 - e. Documents
 - f. Locations

Exercise 1 - The External Perspective

Situation:

The strategic planning team of DAC is getting ready to develop the business plan for next year. They are looking at the external environment to see what is needed in responding to changes in the business landscape. They have decided to do a PESTLE – M analysis and see what issues they are facing. Along with the PESLE – M work they are using the Blue Ocean/ Red Ocean (BO/RO) perspective to assess what moves are appropriate. They are working with a consulting firm that has created a description of the BO/RO approach.

A Bit of Background:

“Red Oceans are all the industries in existence today – the known market space. In the red oceans, industry boundaries are defined and accepted, and the competitive rules of the game are known. Here companies try to outperform their rivals to grab a greater share of product or service demand. As the market space gets crowded, prospects for profits and growth are reduced. Products become commodities or niche, and cutthroat competition turns the ocean bloody. Hence, the term of *red oceans* is used.

Blue oceans, in contrast, denote all the industries not in existence today – the unknown market space, untainted by competition. In blue oceans, demand is created rather than fought over. There is ample opportunity for growth that is both profitable and rapid. In blue oceans, competition is irrelevant because the rules of the game are waiting to be set. Blue Ocean is an analogy to describe the wider, deeper potential of market space that is not yet explored. The cornerstone of Blue Ocean Strategy is 'Value Innovation'. A blue ocean is created when a company achieves value innovation that creates value simultaneously for both the buyer and the company. The *innovation* (in product, service, or delivery) must raise and create value for the market, while simultaneously reducing or eliminating features or services that are less valued by the current or future market.”¹

In summary, a Blue Ocean strategy is where demand is created rather than fought over. There is ample opportunity for growth that is both profitable and rapid. EBay, Cirque du Soleil, and Dubai UAE’s free world-class trade zones are examples of Blue Ocean strategy rolled out.

A recent change has been suggested by the authors, W. Chan Kim and Renee Mauborgne in recent Harvard Business Review article is to extend the analysis using a graphic like this. ²

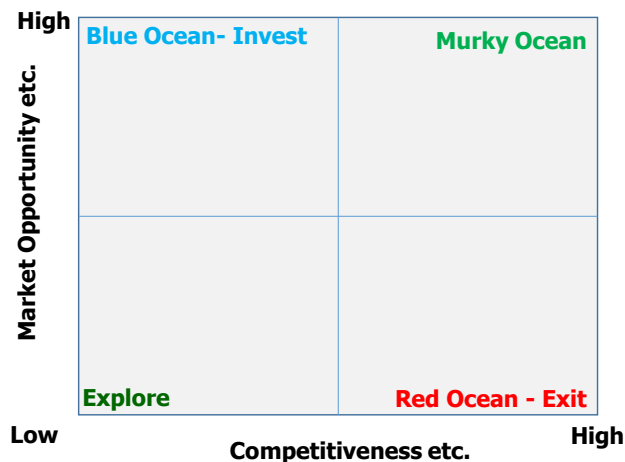
¹ http://en.wikipedia.org/wiki/Blue_Ocean_Strategy

² “How Strategy Shapes Structure”, W. Chan Kim and Renee Mauborgne, Harvard Business Review, September, 2009

The Blue Ocean Approach:

- A blue ocean is the one in which there are few competitors and you have a hot product
- A red ocean is where there are lots of products and lots of competitors and you are fighting for market share
- A simple graphical way to see this relationship is the 4-Box Approach

The 4 – box approach



What you do:

We are looking for a hint at strategic direction based on the business landscape. First look at the ranking of the markets below based on the environment defined by the composite view of PESTLE and the attributes of the markets. Next look at the filled in 4-box below for DAC. Looks at where all the dots are. Here are some assertions about what you see:

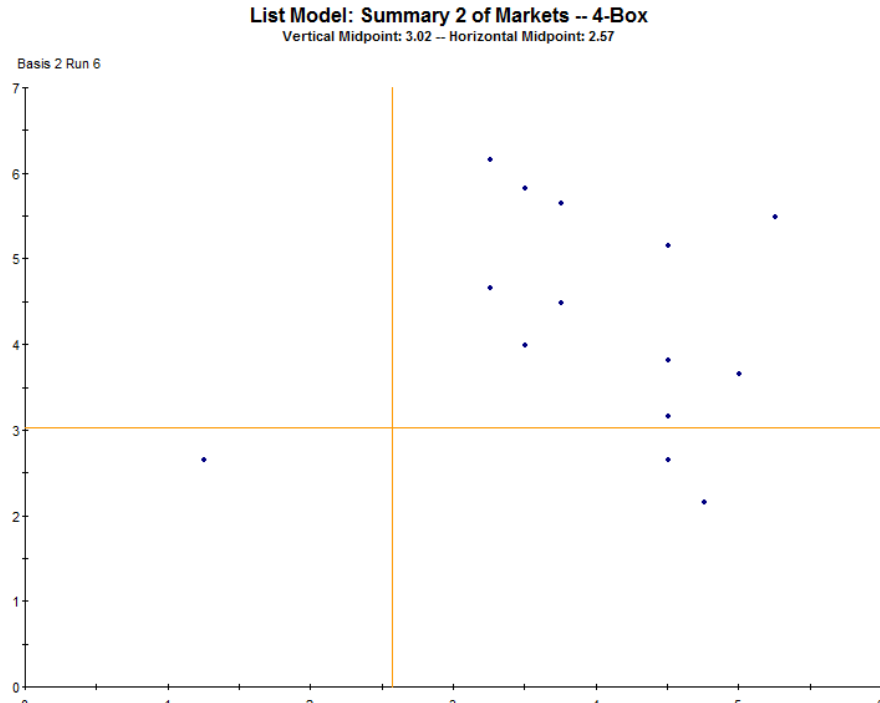
From Column 1 – The Home laundry product market looks like the best place to be. This is a rank by best market on top and worst on the bottom. For example SEA has lots of competitors whereas the home laundry market does not.

From Column 2 - The Americas looks like the best market in terms of PESTLE context. This is one of the largest markets with a limited number of competitors in it. One reason so many foreign firms want to enter this market.

What would you suggest they do?

ATTRIBUTES USED: Market Size Market Share Number of Products Market Potential Market Growth Number of Competitors INSTANCE RANK: 1 -- 2.17 -- Prod - Home laundry 2 -- 2.67 -- Prod - Refrigeration 2 -- 2.67 -- Geo - Americas 3 -- 3.17 -- Prod - Heating 4 -- 3.67 -- Prod - Residentail water Heaters 5 -- 3.83 -- Prod - Gas cooking 6 -- 4. -- Prod - Air Conditioning 7 -- 4.5 -- Demo - Age bracket 8 -- 4.67 -- Demo - Income 9 -- 5.17 -- Prod - Commercial water heaters 10 -- 5.5 -- Prod - Dishwshers 11 -- 5.67 -- Prod - Electronic cooking 12 -- 5.83 -- Geo - EU 13 -- 6.17 -- Geo - MENA 13 -- 6.17 -- Geo - SEA	Markets --> Legislative Matrix) Economic Impact on Markets Markets --> Economic Matrix) Market Impact by Social Markets --> Social Matrix) Technology Impacts on Markets Markets --> Technologies INSTANCE RANK: 1 -- 1.25 -- Geo - Americas 2 -- 3.25 -- Geo - SEA 2 -- 3.25 -- Geo - MENA 2 -- 3.25 -- Demo - Income 3 -- 3.5 -- Prod - Air Conditioning 3 -- 3.5 -- Geo - EU 4 -- 3.75 -- Prod - Electronic cooking 4 -- 3.75 -- Demo - Age bracket 5 -- 4.5 -- Prod - Refrigeration 5 -- 4.5 -- Prod - Commercial water heaters 5 -- 4.5 -- Prod - Heating 5 -- 4.5 -- Prod - Gas cooking 6 -- 4.75 -- Prod - Home laundry 7 -- 5. -- Prod - Residentail water Heaters 8 -- 5.25 -- Prod - Dishwshers
--	--

Most of the DAC dots are in the upper right quadrant.



Those who have Gadrian active and installed on their machine can look at specific rankings of the market opportunities. Once done, use the following steps:

1. Click on the B in the upper left, this takes you to the browser
2. Click on the plus sign (+) by the list models
3. Find the Market List model and click on the name
4. The model is displayed to the right and you can see the attributes of each market

What can you learn from looking at the individual attributes?

Exercise 2 – External Architecture Impact on Strategic Architecture

Situation:

DAC management is considering some key changes in the strategies of the business. They would like to know if they change a market what capabilities would be impacted the most. When they find that out they can decide which paths to evaluate using some form of decision analysis to evaluate alternatives using quantitative methods. This sounds like a simple question since there are no quantities involved. In reality several steps are needed plus there is some qualitative material that must be analyzed.

The qualitative material is usually in the form of interviews that get summarized into relationship matrices. In this exercise the material has already been prepared. This is an alignment issue. You need to know the alignment of capabilities with markets to assess impact. Usually the direct alignment of capabilities and markets is not available. However, often there may be other material that can be linked to get the same result.

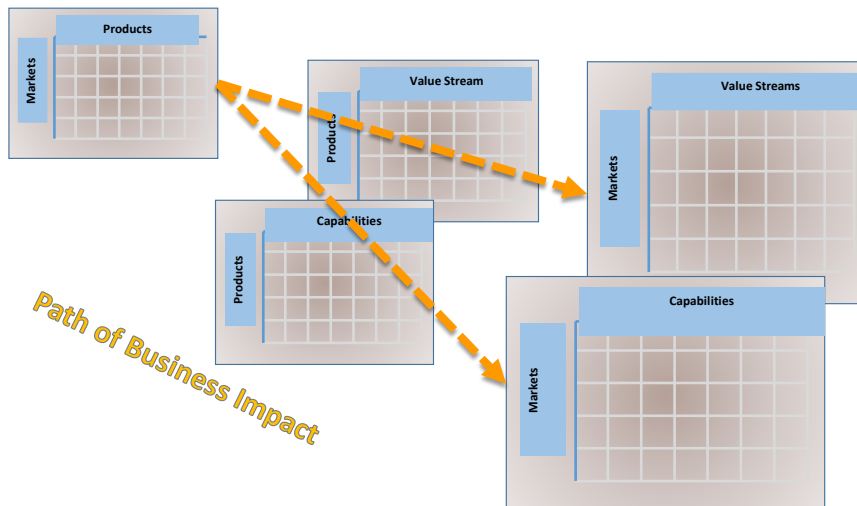
To solve this linking problem we use an inferencing technique on material that is typically available from studies that were done on strategic direction of the business. The alignment uses 2 matrices, one that links markets to products and another that links products to capabilities. The markets form the row headings and the products form the column headings. A second matrix relates the products as row headings to the capabilities as column headings. There are other combinations that can be used for this and it depends on the type of study, the scope and the level of detail that was done and the matrices that were developed. Also, this inferencing technique is general and can be applied to any number of alignment problems.

The question being answered is ‘What capabilities might be impacted if a particular market is changed?’ The alignment of the 2 matrices is used to assess this impact. You can also use this to rank the capabilities based on strategic impact where the question is ‘What capabilities are impacted by the most markets?’ This tells you that a change in any of several markets will impact a particular capability and hence it is sensitive to any market change.

For a more complete description of the inferencing technique see the lecture material.

Key Linkage of Markets to Capabilities and Value Streams

Exercise 2

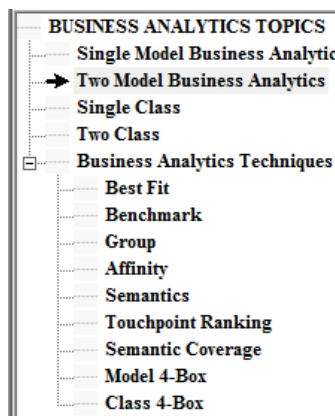


Choose one path
to work with, in
this case
capabilities

What you do:

The steps below assume Gadrian is open and active on the desktop using the file “Workshop DAC Analytics Ver 5.1 June 2015”.

1. Open Gadrian to the **Analytic Browser**.
2. Select the “**2 Model Analytics**” workspace.



3. Select “**Matrix**” for the Model Type analytic
4. There are 2 list boxes that will display the compatible models for matrix analytics.

5. Select the “**Market to Product**” model in the Available A Matrix Models (left box) and then the name will be shown below under the Matrix Model A Name box below.
6. Select the “**Product to Capability**” model in the Available B Matrix Models (right box) and then the name will be shown below under the Matrix Model B Name box below.
7. Click on the  symbol to execute the inference. The results are shown in the display area at the lower part of the screen with all the tabs. Note that it says to Save the model use the File menu.
8. Click on the File menu item and select “**Save Inference as Model**”, a pop-up with an input field will open with the heading ‘Save Inference Model’ just above the results area.
9. Click in the space and type a name for the inferred model, call it ‘**Market Impact on Capability**’ and click on Save.
10. A dialog box opens and asks if you want to create a matrix model by that name showing **67** relations and you click **Yes**.
11. A message appears that says if the model was created successfully. Click on **OK**.
12. You can see the frequencies by going to the **Analytics Browser** and selecting the *Affinity Technique* workspace and doing the following:
 - a. Click on the Available Models drop down arrow
 - b. Select the Class Relations named “**Market Impact on Capabilities**”
 - c. The Affinity Actions list box becomes active
 - d. Select **Frequencies** and the Row and Column frequencies are displayed
13. There are 2 core questions to answer:
 - a. What is the most frequently linked product (largest row number)
 - b. What is the most frequently linked capability (largest column number)
14. How do you know this?
 - a. Sum the dots in each row and column to answer these questions for you.
 - b. Place the numbers in the boxes below focusing on the highest numbers.
 - c. Write the name and number of dots for the highest (most frequently referenced) 3 rows and columns in the tables below:

	Name of market (row headings)	Number of capabilities (dots)
1		
2		
3		

	Name of capability (column headings)	Number of markets (dots)

1		
2		
3		

15. **Lesson learned** - When looking at the top 3 or 4 frequencies in *markets* and *capabilities* you see that:

- For *capabilities* (column numbers), these are the most likely to be impacted by a change in the markets of the business. In this case, the top 2 capabilities are impacted by at least X of the (total) markets so a change in any of those X markets will impact these Y capabilities.
- For *markets* (row numbers), these are the most likely to have the greatest impact on capabilities. In this case, if you change the top 2 ranked markets as many as Y capabilities are impacted to some degree for each product.
- This identifies the potential risk to capabilities when markets change.
- The degree of change has not been identified at this point. The goal is to identify what is impacted first. Then the degree can be established.
- We can do the same for *value streams*.

Markets		Capabilities	
Affinity Workspace			
Row Frequencies		Column Frequencies	
Instance Name	Frequency	Instance Name	Frequency
Geo - SEA	9	Financial Management	12
Geo - MENA	9	Vendor Management	12
Geo - EU	9	Customer Service Management	12
Geo - Americas	9	Manage Product Acquisition	11
Demo - Income	9	Customer Relationship Management	11
Demo - Age bracket	9	Personnel management	10
Prod - Refrigeration	7	Manufacturing Management	10
Prod - Electronic cooking	7	Product Engineering Management	10
Prod - Heating	6	Product Design Management	10
Prod - Commercial water heaters	5		
Prod - Air Conditioning	5		
Prod - Home laundry	5		
Prod - Gas cooking	5		
Prod - Residential water Heaters	2		
Prod - Dishwashers	2		

Below is the matrix that generated the affinity frequencies.

All
Expand Contract Class Relations: 98 -- Density: 65.33%

From	To	Customer Relationship Management	Customer Service Management	Financial Management	Manage Product Acquisition	Manufacturing Management	Personnel management	Product Design Management	Product Engineering	Vendor Management
Demo - Age bracket		●	●	●	●	●	●	●	●	●
Demo - Income (M		●	●	●	●	●	●	●	●	●
Geo - Americas (M		●	●	●	●	●	●	●	●	●
Geo - EU (Market		●	●	●	●	●	●	●	●	●
Geo - MENA (Mar		●	●	●	●	●	●	●	●	●
Geo - SEA (Marke		●	●	●	●	●	●	●	●	●
Prod - Air Condition			●			●	●	●	●	
Prod - Commercial		●	●	●	●					●
Prod - Dishwashers				●						●
Prod - Electronic c		●	●		●	●		●	●	
Prod - Gas cookin		●	●	●	●					●
Prod - Heating (M				●		●	●	●	●	●
Prod - Refrigeratio		●	●		●	●	●	●	●	
Prod - Residential				●						●

The value stream version:

Markets	Value Streams																																																						
<i>Affinity Workspace</i>																																																							
<table> <tr> <th colspan="2">Row Frequencies</th></tr> <tr> <th>Instance Name</th><th>Frequency</th></tr> <tr><td>Prod - Heating</td><td>8</td></tr> <tr><td>Geo - SEA</td><td>8</td></tr> <tr><td>Geo - MENA</td><td>8</td></tr> <tr><td>Geo - EU</td><td>8</td></tr> <tr><td>Geo - Americas</td><td>8</td></tr> <tr><td>Demo - Income</td><td>8</td></tr> <tr><td>Demo - Age bracket</td><td>8</td></tr> <tr><td>Prod - Air Conditioning</td><td>7</td></tr> <tr><td>Prod - Refrigeration</td><td>7</td></tr> <tr><td>Prod - Electronic cooking</td><td>7</td></tr> <tr><td>d - Commercial water heaters</td><td>1</td></tr> <tr><td>d - Residential water Heaters</td><td>1</td></tr> <tr><td>Prod - Dishwashers</td><td>1</td></tr> <tr><td>Prod - Home laundry</td><td>1</td></tr> <tr><td>Prod - Gas cooking</td><td>1</td></tr> </table>	Row Frequencies		Instance Name	Frequency	Prod - Heating	8	Geo - SEA	8	Geo - MENA	8	Geo - EU	8	Geo - Americas	8	Demo - Income	8	Demo - Age bracket	8	Prod - Air Conditioning	7	Prod - Refrigeration	7	Prod - Electronic cooking	7	d - Commercial water heaters	1	d - Residential water Heaters	1	Prod - Dishwashers	1	Prod - Home laundry	1	Prod - Gas cooking	1	<table> <tr> <th colspan="2">Column Frequencies</th></tr> <tr> <th>Instance Name</th><th>Frequency</th></tr> <tr><td>Manage Supply Chain</td><td>12</td></tr> <tr><td>Build Products</td><td>10</td></tr> <tr><td>Select Suppliers</td><td>10</td></tr> <tr><td>Manage Information</td><td>10</td></tr> <tr><td>Ship to Customer</td><td>10</td></tr> <tr><td>Supply Parts</td><td>10</td></tr> <tr><td>Service Customers</td><td>10</td></tr> <tr><td>Maintain Products</td><td>10</td></tr> </table>	Column Frequencies		Instance Name	Frequency	Manage Supply Chain	12	Build Products	10	Select Suppliers	10	Manage Information	10	Ship to Customer	10	Supply Parts	10	Service Customers	10	Maintain Products	10
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Time Allowed: Total Time is 30 Minutes

- 15 minutes to do the analytic
- 15 minutes for team and group discussion and

Discussing the results

Team Discussion topic (15 Minutes):

- Where would you use this type of impact analysis in your organization?
- List 3 to 5 examples for use on the flip charts
- Each team will have one person describe the uses

Exercise 3 – Business Architecture impact on Enterprise Architecture

Aligning capabilities with processes, information and applications

Situation:

Understanding the impact on the operational components of the organization require one more inference. The final inference will link the external architecture of markets to 3 key parts of the organization namely, applications, processes and data. While not perfect, the inference technique based on the semantics of the names of business components points to the direction for possible impact. Because of terminology differences, there will be different naming conventions or using the same words for something that is different between operating units so some cleanup of the name is very useful.

However, if the cultural differences are not great then the terminology is likely to be similar. This makes basic semantic phrase analysis very useful in such situations. Also, the use of standard industry terminology, software packages, the Internet, standardized requirements and other shared services has made terminology more common.

The degree of impact gives an indication of how difficult change may be or how severe the change may be between the changes in the market and the execution response to management direction. Think of management direction as a filter for the market changes.

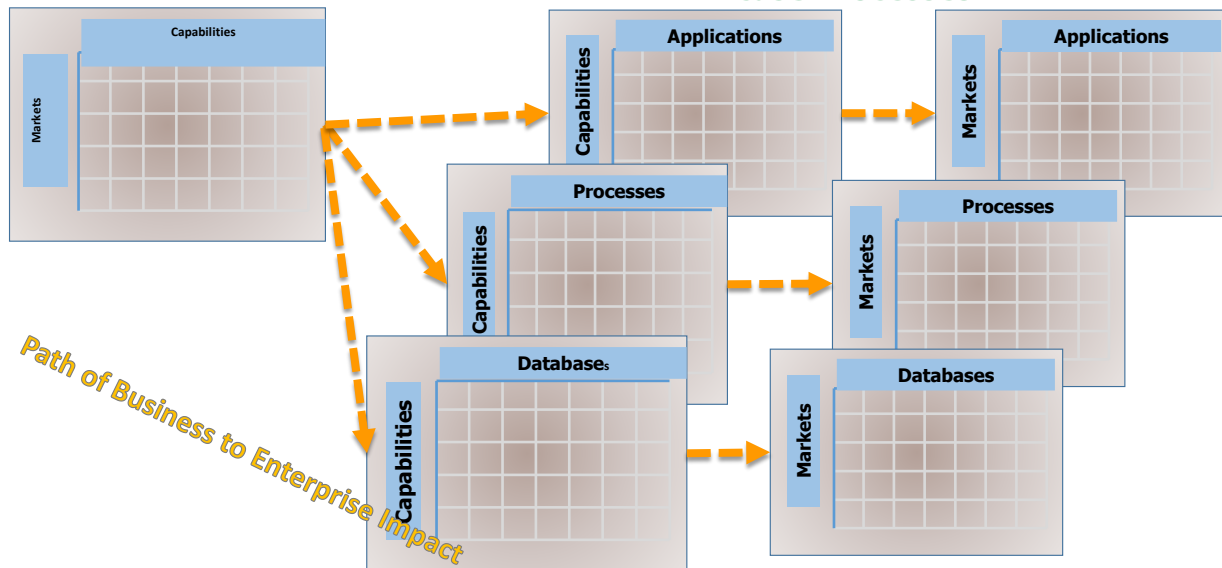
What You Have:

The inference idea is basic. There are added variations of the inference that provide a more specific and detail idea of impact but for now this basic level is sufficient as it point to where the changes have impact.

Key Linkage of Markets to Applications, Databases and Processes

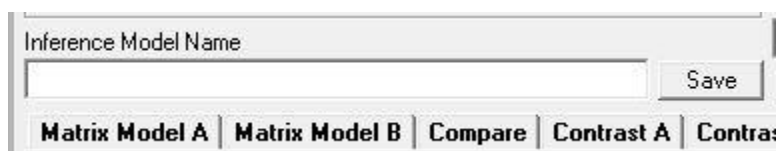
Exercise 3

Choose one path to work with, in this case Processes



What you do- Tool Analysis 20 Minutes:

1. Open up the Gadrian Analytics browser
2. Select the 2 model analytics workspace.
3. Select the *Matrix* radio button for the model type to use for the analytic
4. There are 2 list boxes that will display the compatible models for matrix analytics.
5. Select the “**Market to Capabilities**” model in the left box.
6. Select the operating unit “**Capabilities to Processes**” model in the right box.
7. Click on the  symbol to execute the inference. The results are shown in the display area at the lower part of the screen with all the tabs.
8. Click on the Infer tab and look at the number of instances that are inferred.
9. Click on the ‘File’ menu item
10. Click on the Save Inference as Model option



11. Enter the name of the model in the field that is opened

12. The matrix should look like the one below.

Model Name: Company A Inferred Market to Processes [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: FUNCTIONS]

All
☐ Expand ☒ Contract Class Relations: 93 -- Density: 18.24%

From	To	Build Appliances (DAC Core)	Develop Human Resources (DAC Core)	Develop New Products (DAC Core)	Manage Quality (DAC Core)	Manage Supply Chain (DAC Core)	Market Appliances (DAC Core)	Report on Financial Results (DAC Core)	Research and Development (DAC Core)
Demo - Age brack		●	●	●	●	●	●	●	●
Demo - Income (M		●	●	●	●	●	●	●	●
Geo - Americas (M		●	●	●	●	●	●	●	●
Geo - EU (Market		●	●	●	●	●	●	●	●
Geo - MENA (Mar		●	●	●	●	●	●	●	●
Geo - SEA (Marke		●	●	●	●	●	●	●	●
Prod - Air Condition		●	●	●	●	●			●
Prod - Commercial		●				●	●	●	
Prod - Dishwashers		●				●		●	
Prod - Electronic c		●	●	●	●	●	●		●
Prod - Gas cookin		●				●	●	●	
Prod - Heating (M		●	●	●	●	●		●	●
Prod - Refrigeratio		●	●	●	●	●	●		●
Prod - Residential		●				●		●	

Markets > DAC Core Processes

16. Click on the Analytics workspace

17. Select Affinity from the techniques

18. Click the drop down and look at the Model Selection window, it should look like this:

Available Models

Inferred Impact of External Change on Processes

Model Name: Inferred Impact of External Change on Processes

Row Dimension: EXTERNAL ARCHITECTURE

Column Dimension: FUNCTIONS

Class Relations

Legislative --> DAC Core Processes

19. Locate and click on the model that says 'Company A Inferred Market to Process' you should see a screen with options at the bottom like this:

Row Class: Legislative

Column Class: DAC Core Processes

Matrix Density: 50.27%

Affinity Actions

☐ Connections Grid
☐ Frequencies
☐ Row Affinities
☐ Column Affinities
☐ Skew Factors

20. Click the frequency options and you will see a screen that looks like the one below. This is what you discuss.

21. Look at the most frequently referenced market (rows) and most frequently referenced process (Columns), what do think when you see this?

<i>Affinity Workspace</i>			
Row Frequencies		Column Frequencies	
Instance Name	Frequency	Instance Name	Frequency
Geo - SEA	8	Manage Supply Chain	15
Geo - MENA	8	Build Appliances	15
Geo - EU	8	Report on Financial Results	12
Geo - Americas	8	Market Appliances	11
Demo - Income	8	earch and Develop Products	10
Demo - Age bracket	8	Develop Human Resources	10
Prod - Heating	7	Manage Quality	10
Prod - Refrigeration	7	Develop New Products	10
Prod - Electronic cooking	7		
Prod - Air Conditioning	6		
d - Commercial water heaters	4		
Prod- Home laundry	4		
Prod - Gas cooking	4		
d - Residentail water Heaters	3		
Prod - Dishwshers	3		

Questions about the problem:

- Assuming this works what would you conclude about the processes and markets?
- What if the there are multiple market scenarios?
- Is the granularity or is leveling good enough for decisions or planning?
- Would you analyze multiple scenarios and choose the best? What if management dictates the scenario?

Time Allowed: Total Time is 30 Minutes